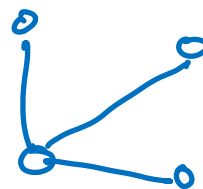


Lecture 13.



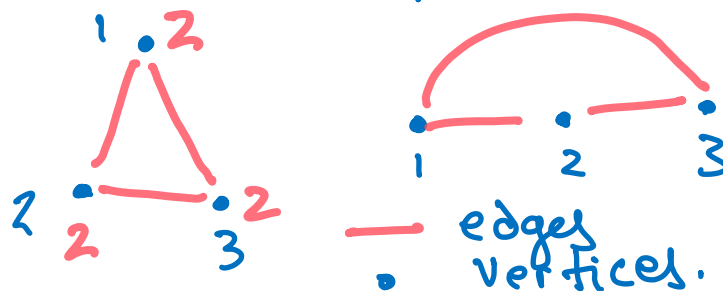
Graph Theory:

Graph: Mathematical abstraction of pairwise relationships.

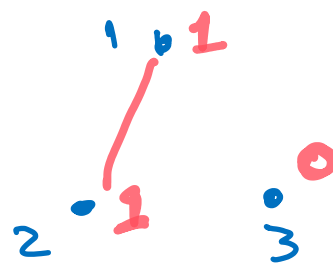
Defn: A ^(multi) graph $G = (V, E)$ is a pair of sets V , called the set of vertices, E , called the set ^(multi) of edges, s.t. E is a ^(multi) set of unordered pairs of elements of V , i.e. $E \subseteq \{S \subseteq V : |S|=2\}$.

ex 1: $V = \{1, 2, 3\}$
 $E = \{\{1, 2\}, \{2, 3\}, \{1, 3\}\}$

Drawings



ex 2: $V = \{1, 2, 3\}$
 $E = \{ \{1, 2\} \}$

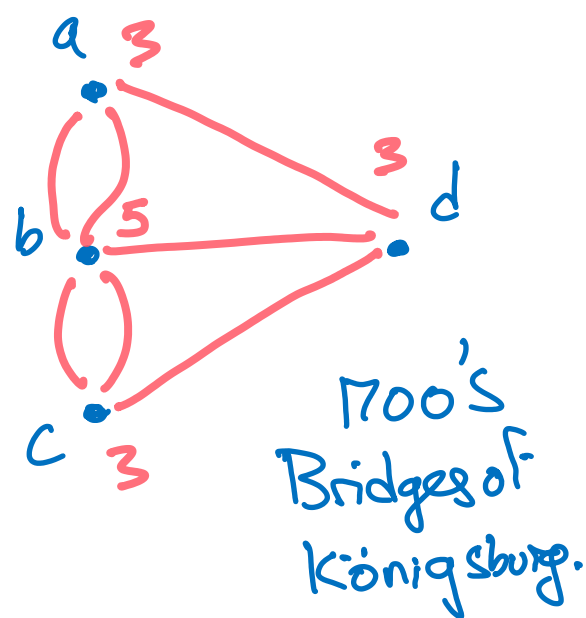


ex 3: $V = \{a, b, c, d\}$
 $E = \{ ad, bd, cd, \underline{ab}, \underline{ab}, \underline{bc}, \underline{bc} \}$

dropped $\{, \}$
 shorthand

repeated edges

multiset.



Remark: If E has repetitions, G is a multigraph.
 otherwise, G is a simple graph.

- No directed graphs.
- All graphs are finite in 5B.

Why?

① Internet.

② Social Network.

③ Brain, Neural Network, Molecules, Feynman Diagram.

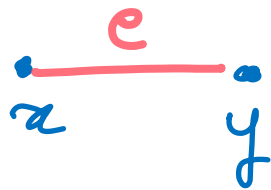
④ Math

② Degree and Handshaking

III Connectivity

Terminology: Given $G=(V,E)$, if $e=xy \in E$ then

x is adjacent to y in G



e is incident with x in G
" " y in G .

x, y are the endpoints of e in G .

Defn: Given $G=(V,E)$ the degree of $v \in V$ in G is

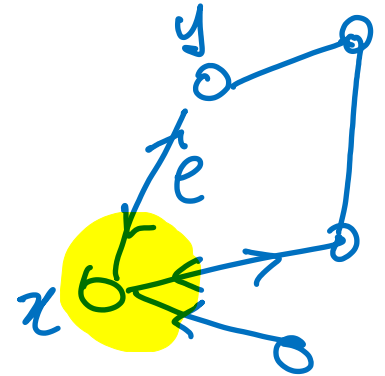
$$\deg(v) = |\{e \in E : e \text{ is incident with } v \text{ in } G\}|$$

Theorem (Handshaking) If $G=(V,E)$ is a graph then

If $V=\{v_1, \dots, v_n\}$

$\deg(v_1) + \deg(v_2) + \dots + \deg(v_n)$

$$\sum_{v \in V} \deg(v) = 2|E|$$



Proof: Assume $G=(V,E)$ is arbitrary.

Consider the set of incidences

$$I = \{ (e, v) : e \text{ is incident with } v \text{ in } G \}$$

"handshake impressions"

We will count the number of incidences $|I|$ in two different ways:

- (i) Each edge $e=xy \in E$ participates in exactly two incidences (e,x) and (e,y) . No incidence appears in more than one edge. Thus, $|I|=2|E|$.

(edge-wise)

(vertex-wise) (2) Each vertex $v \in V$ participates in $\deg(v)$ incidences. No incidence participates in more than one vertex. Thus,

$$|I| = \sum_{v \in V} \deg(v).$$

We conclude that $\sum_{v \in V} \deg(v) = 2|E|$, as desired. $\square$

Pol1: $\exists?$ G degrees $1, 2, 3, 3, 4$.

No: $\sum_{v \in G} \deg(v) = 13$, not even, would violate handshaking.

Corollary: In any G , the number of odd degree vertex is even.

Proof: Let G graph, n_1 be the # of odd degree vertices
 n_2 even.

Then: $0 \equiv \sum_{v \in V} \deg(v) \pmod{2}$ (mod 2) by handshaking

$$\equiv n_1 \cdot 1 + n_2 \cdot 0 \pmod{2}$$
$$\equiv n_1 \pmod{2}.$$

□

Ramsey Theory Any large structure must contain patterns.

Theorem: In any $\forall$ group of 6 people, there $\exists$
are at least 3 mutual acquaintances Δ
or 3 mutual strangers. $\therefore$

Proof: Consider the graph $G=(V,E)$ defined by

models pairwise relationships

$$V = \{1, 2, 3, 4, 5, 6\}.$$

$$E = \{xy : x \text{ and } y \text{ are acquaintances}\}.$$

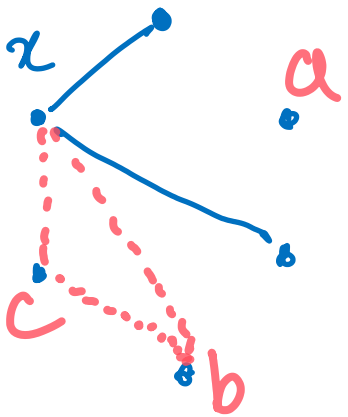
Choose an arbitrary $x \in V$.

Case 1: $\deg(x) \leq 2$: Since there are 5 vertices in $V - \{x\}$, we can choose three distinct $a, b, c \in V$ such that

$xa, xb, xc \notin E$. Observe that either

- $\{ab, bc, ca\} \subseteq E$, in which case a, b, c are mutual acquaintances.

- Without loss of generality, $bc \notin E$, so x, b, c are mutual strangers.

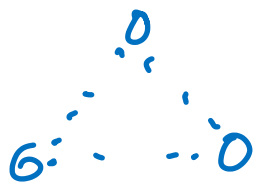


Case 2: $\deg(x) \geq 3$: $\langle Hw \rangle$.

□

Thm: $\forall$ group of 18 $\exists$ 4 mutual acquaintances or strangers.

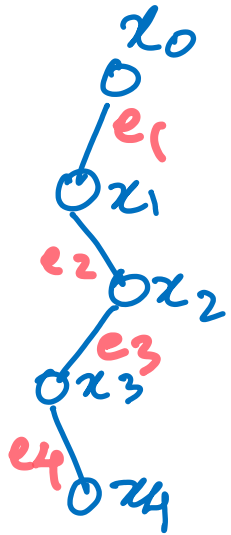
Ramsey: $\forall k \geq 2 \exists N$ s.t. $\forall$ group of N people
 $\exists$ subgroup of k
 mutual strangers or acquaintances.



k	N
3	6
4	18
5	43-46
$\vdots$	
k	$N \leq 4^k$ [Sah'24].

III Connectivity

Defn: A path of length k in a graph $G=(V,E)$ is an alternating sequence of vertices $x_i \in V$ and edges $e_i \in E$:



$$x_0, e_1, x_1, e_2, \dots, e_k, x_k$$

Such that for every $i=1 \dots k$ $e_i = \{x_{i-1}, x_i\}$.

A path with no repeated vertices is called simple.

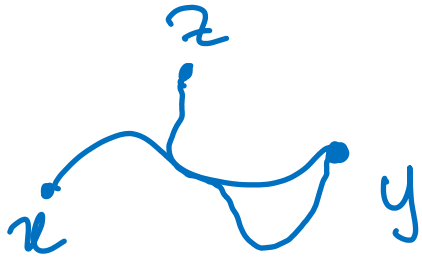
Remark: • If $x \overset{G}{\rightsquigarrow} y$ then $y \overset{G}{\rightsquigarrow} x$ there is a path from y to x in G .

$x, e_1, \dots, e_k, y$ can be reversed to get
 $y, e_k, \dots, e_1, x$.

- If $x \overset{G}{\rightsquigarrow} y$ and $y \overset{G}{\rightsquigarrow} z$ then
 $x \overset{G}{\rightsquigarrow} z$.

Can concatenate $x, e_1, \dots, e_k, y$ and $y, f_1, \dots, f_\ell, z$
 to get $x, e_1, \dots, e_k, y, f_1, \dots, f_\ell, z$.

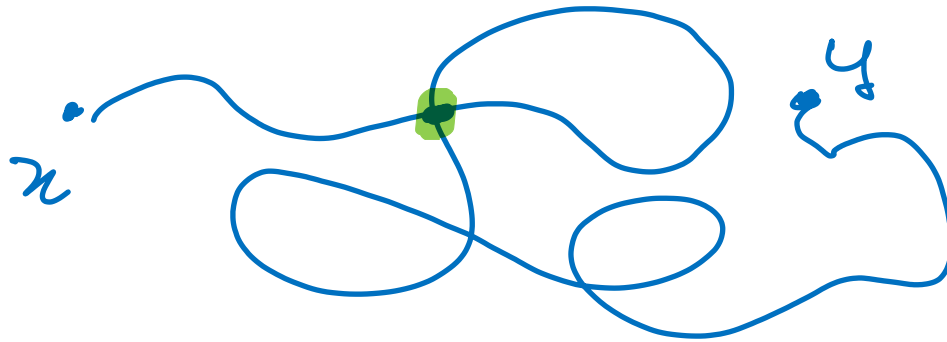
Q)



ok to reuse vertices, edges.

Proposition: If there is a path between $x \neq y$ in G ,
 then there is a simple path between x and y
 in G .

Idea:



repeated
vertices
correspond to
"loops".

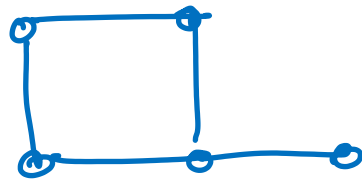
Sketch: Let $\gamma = x, e_1, x_1, e_2, \dots, y$

be a path of shortest length from x to y in G .

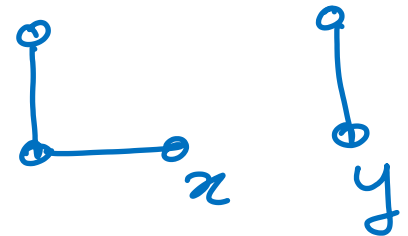
If there is a repeated vertex w , can remove a loop to get a shorter path. $\square$
<HW>.

Defn: A graph $G = (V, E)$ is connected if for every $x, y \in V$ there is a path from x to y in G .

ex 5:



connected



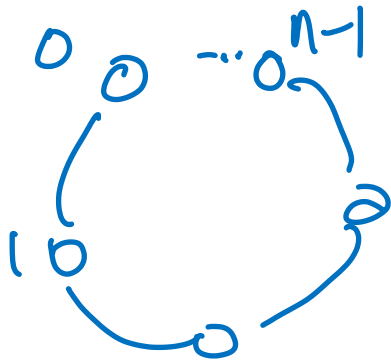
not connected.

ex 6: (Cycle) if $n \in \mathbb{Z}_+$, define

$$C_n = (V, E)$$

$$V = \{0, 1, \dots, n-1\}$$

$$E = \{xy : x - y \equiv 1 \pmod{n}\}.$$



HW: Show this is connected.

AI/Math:

Feb 5: Posted 10 problems on internet.

Feb 14: Solns posted.

+ 2/10 solved by lots of people.

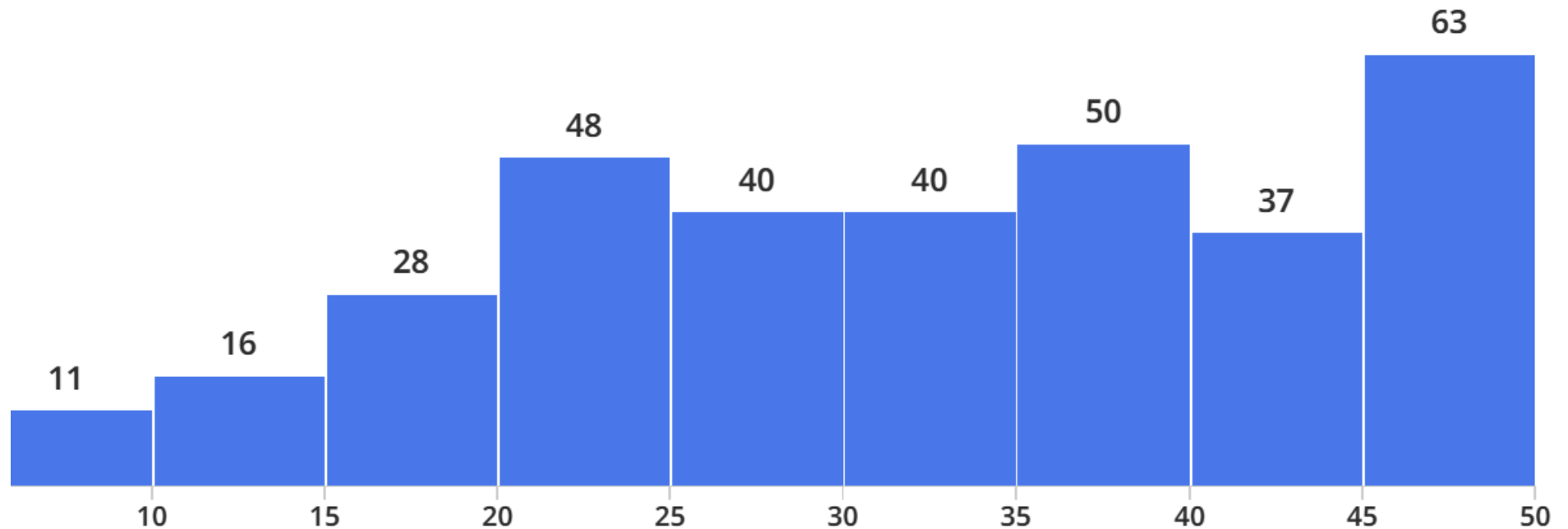
+ 3-5 additional were solved by AI/CDM.

+ Very hard to verify correctness.

Grades for Midterm 1

● Regrade Requests Closed

● Grades Published



Median

32.5

Maximum

50.0

Mean

32.01

Std Dev [?](#)

11.57